

Overview

Topic 1 introduces carbon farming, carbon farming activities, and their purposes, which include:

- reducing emissions
- storing more carbon
- delivering economic and other co-benefits to farmers and land managers, the environment and communities.

This topic introduces key concepts from the farmer's and land manager's viewpoint: how increasing carbon storage on their land and reducing emissions from their operations helps reduce their carbon footprint. It introduces carbon accounting, tools and calculators to estimate a carbon footprint.

The topic explains how Aboriginal and Torres Strait Islander people (Australia's First Peoples, referred to in this program as First Nations people) have cared for Country for over 60,000 years and that carbon farming activities can potentially align First Nations peoples' traditional knowledge and recent science-based farming and land management methods.

The topic drills down into increasing carbon storage (sequestration) by explaining the carbon cycle, soil organic carbon, the greenhouse effect, and climate change. It examines the agriculture sector's emissions profile, which is mostly methane and nitrous oxide emissions.

The topic concludes by explaining 'carbon neutral', 'net zero emissions' and possible pathways for action — earning Australian Carbon Credit Units (ACCUs) under the Australian Carbon Credit Unit Scheme (ACCU Scheme) and gaining Climate Active certification — while recognising some farmers and land managers will choose to do neither.

Learning outcomes

After completing this topic, you will be able to:

- describe carbon farming and carbon farming activities
- classify greenhouse gas (GHG) emissions and carbon sinks relevant to agriculture and land management
- explain how carbon farming can potentially align First Nations peoples' traditional knowledge with recent science based farming and land management methods
- acknowledge the carbon cycle and nitrogen cycle, their relationship to GHGs, and their role in climate change
- outline key carbon farming-related concepts, including carbon neutrality, net zero emissions, Australian Carbon Credit Units (ACCUs), and Climate Active certification.

Watch these videos

In this video (3:59 minutes), presenters Gail Reynolds-Adamson and Matt Woods introduce Topic 1 and provide important context.

GAIL REYNOLDS-ADAMSON: In this first topic, we'll look at the basics of carbon farming, what it is, and why we need it. There are many different reasons why you might choose to carbon farm, and we will cover these benefits in the next topic.

But there's one reason why carbon farming is important, and it affects all of us. Yep. Climate change.

MATT WOODS: Now we know that we don't need to tell you how a warming planet, because of carbon and other emissions in the atmosphere, is playing havoc with our climate.

As farmers, you're experiencing firsthand increasingly severe and frequent extreme weather events like drought, fire, and floods that are negatively affecting your livelihoods.

The hard reality is that farming is part of the problem, with agriculture accounting for about 17 per cent of total greenhouse gas emissions in Australia.

The good news is that farmers can also be a big part of the solution, and carbon farming helps with this.

REYNOLDS-ADAMSON: Aboriginal and Torres Strait Islander people, like all First Nations people around the world, have been caring for our land for tens of thousands of years. We are connected to Country. We are part of the Country, and Country is part of us.

We take a holistic approach and acknowledge the importance of sustainable land management, emphasising health of our land, the well-being of the ecosystem, and the preservation of cultural heritage.

In short, through traditional land management practices, we care for Country and Country cares for us.

Of course, it's not only First Nations people who have a connection to this land and a desire to care for it. The beauty of carbon farming is it allows us to combine the best of both worlds, applying ancient cultural practices such as savanna burning with the Western scientific knowledge system.

Modern day technologies can help Indigenous and non-Indigenous farmers alike to reduce carbon emissions.

We've learned in our tree rejuvenation project there are different approaches that you can use to gain both financial and non-financial benefits.

WOODS: Carbon farming is also usually good for business.

Sequestering carbon and reducing greenhouse gas emissions can increase profitability.

Climate change and carbon credits aside, carbon farming can make sense for farmers and land managers if they want to improve their bottom line and the environment.

Selling carbon credits is a further option for land managers and farmers.

However, making money from carbon credits is complex and, like most ventures, has risks.

In this topic, we'll look at some basic climate science and the impact of agricultural greenhouse gas emissions.

We'll also explore carbon farming in more detail and the types of activities you can do to reduce greenhouse gas emissions in your sector.

In addition, we'll see some traditional First Nations practices, and finally, look at the difference between carbon credits and carbon neutral accreditation.

REYNOLDS-ADAMSON: We all have a connection to this land. For Aboriginal and Torres Strait Islander people, we know this is caring for Country, which we have done since time immemorial.

Now with climate change, we need to look at repairing Country for future generations.

Mother Earth does not see colour.

Collectively, we have a responsibility to repair our environment.

Let us share our knowledge and experiences to support each other to have a mutually beneficial outcome and leave this Country better than how we received it.

In this video (2:13 minutes), Matt Woods and Professor Richard Eckard of the University of Melbourne discuss carbon farming in relation to carbon capture in plants through photosynthesis.

MATT WOODS: Hello. I'm Matt Woods, and I'm here with Professor Richard Eckard. Richard has been working for over 20 years on addressing the impacts of a changing climate on agriculture.

How would you describe what carbon farming actually is?

PROFESSOR RICHARD ECKARD: Well, ironically, carbon farming all farmers are in the carbon business because what most people don't realise is that almost half of all organic vegetation, this is wheat, pastures, any crops that we produce, almost half of it is carbon. So, actually, we're in the game of carbon conversion. We capture carbon out of the atmosphere through photosynthesis.

It takes CO₂ out of the atmosphere, carbon dioxide, locks it into plants in carbohydrates. We either feed that to humans or feed it to animals. So, actually, all farmers are in the carbon business. We've just never asked the question, how do you be more efficient at that conversion process? How do we capture more carbon out the atmosphere into a better wheat crop, into a better pasture?

And then how efficient are we at converting that into a product that goes through to humans?

But also how are we efficient at capturing or leaving excess carbon in the soil to end up with a greater soil health?

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/training-package/topic-1/1-overview-and-learning-outcomes>